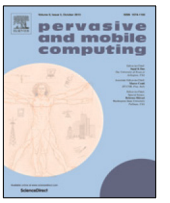
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De$\mathcal{H}$onk: A deep learning based system to characterize vehicular honks in presence of ambient noise



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ABSTRACT

Noise pollution, a growing problem in urban cities, causes severe and long-term health problems. Vehicles on the road and ceaseless honking add to incessant vehicular assault and are the main reasons behind increasing noise pollution. However, analyzing and characterization of honking is not an easy task, and it varies with road traffic, vehicle types, speed, etc. In this paper, we propose a framework De$\mathcal{H}$onk that detects vehicular honks from traffic noise and analyzes various honking features to build a honking profile using only audio samples, which can be further used to infer the context of a particular location, design micro-services, etc. However, the existing methods used only fast fourier transform(FFT) or mel-frequency cepstral coefficients(MFCC) in honk identification, the accuracy lies between 40% to 60% in the presence of ambient noise. Similarly, we failed to obtain good accuracy in recognizing honks at noisy environments by applying existing sound classification models. In the proposed work, we have modeled the sound signal as a spectrogram image to detect vehicular honks. We have collected audio data by traveling 461km of road from various spatio-temporal locations in Durgapur, a sub-urban city in India. Furthermore, the generated spectrogram images are pre-processed, labeled, and fed into various deep learning models for training and testing. In this study, convolutional neural network(CNN), visual geometry group with a very deep convolutional network(VGGNets), efficient convolutional neural networks for mobile vision applications(MobileNet), residual neural network(ResNets), and Inception V3 models are used to train the system to identify honks and honk related features. Experiment results demonstrate that the ResNet152 model has achieved higher accuracy (97.69%) in honk detection as compared to other models. Furthermore, as a micro-service, a honk-aware route recommendation system is developed based on the derived honking features.

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1. Introduction

Noise pollution – a by-product of urbanization and industrialization – is constantly increasing in developing countries like India. Unrestricted traffic movement and incessant honking are the primary sources of growing noise pollution in urban cities. Driver's impatience, sudden braking, and breaking of traffic rules aggravate honking significantly in traffic. However, the sharp rise in number of vehicles on the road has not been accompanied by a proportional improvement of the public road system, and the ceaseless honking and associated traffic noise cause many health problems. Studies

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Table 1
Accuracy of honk identification using FFT and MFCCs.

Method	Without ambient noise (%)	With ambient noise		
		High traffic (%)	Low traffic (%)	Market place (%)
FFT	>98	<45	<55	<60
MFCCs	>70	<40	<45	<40

published in [1,2] focused on the adverse effects of noise pollution on human health and showed that traffic noise causes hypertension, annoyance, hearing loss, and many other diseases. Other studies found in [3,4] showed that high noise exposure interferes with complex task problems and leads to several diseases. In [5], the authors assessed the noise due to heterogeneous traffic sources and analyzed the impact of honking on it.

It may be noted that vehicular honking patterns and their characteristics (intensity, duration, frequency) may widely vary based on different outdoor aspects like road type (highway/street), vehicle type (heavy/light), vehicle density (congestion/free traffic) in a road, time of the day (busy/non-busy hours), area location (residential/market place), etc. Variation induced by such spatio-temporal aspects makes automatic detection of honk features a challenging problem. However, if one can exploit the inherent variation by precisely detecting vehicle honk features using sophisticated models, interestingly, it may provide a fair indication of several urban attributes like traffic noise, congestion, traffic volume, demographic profile of localities, etc., using passive sensing. The inherent correlations [6,7] between honks and outdoor air quality in certain spatio-temporal settings may further be exploited to have a qualitative estimate of outdoor pollution exposure. Hence precise detection, characterization, and localization of honks might be beneficial in developing novel applications and designing policies to minimize the impacts of honking in our daily life. In literature, there are quite a few works that deal with honk detection [8–10] and identification of honk features [8]. However, it may be noted that the audio signals generated by honks can be significantly affected in the presence of ambient outdoor noise like engine noise, roadside music, any kind of loud sound, human voice, etc. Such ambient noise may often mask the audio signature generated due to honks, thereby affecting the detection accuracy to a large extent. Therefore, the main objective of this work is to investigate: *(a) If honk features can be detected precisely and accurately in the presence of ambient outdoor noise? (b) If we can effectively develop a micro-service that can minimize the ill impact of honks in our daily life?*

Motivation: Analysis and characterization of honks in the presence of ambient noise may be challenging. Works proposed in [8,10] mainly used simple FFT to extract honk features from noise samples. In [8], on-road-sensors were deployed to detect vehicular honks, matching honks, and frequency extraction for speed estimation, while in [10], a smartphone based application was proposed for car honk detection. In another work [9], the authors introduced MFCCs for honk identification. Although, the existing works have identified the vehicular honks effectively in road traffic; but they have failed to achieve a good accuracy when vehicular honks are subjected to ambient noise. As a preliminary experiment, we collected some honk samples using smartphones from different areas such as high road traffic areas, residential areas with low traffic, and marketplaces in the presence/absence of ambient noise to show the usefulness of FFT and MFCCs in honk detection. Table 1 represents the achieved accuracy level using FFT and MFCCs with 500 samples, each with 10 min duration. It is observed from the table that in the absence of ambient noise, the FFT in particular achieves good accuracy in honk identification. But the accuracy falls sharply i.e. below 60%, when the sound data contains various ambient noise. Apart from the above works, a significant amount of work has been conducted for sound classification in general, like identifying sound from air conditioners, gunshots, street music, car honk, children playing, dog barking, drilling, engine idling, siren, etc. These works primarily use different deep learning models [11–18]. Here the researchers used CNN [11], SB-CNN [12], ConvNet [13], and other variations of CNN models [14–18]. The above papers use datasets like ESC-50, ESC-10, and UrbanSound8K [11] for model training. Among these ESC-10 does not provide any honk sample. ESC-50 and UrbanSound8K contain car honk samples of duration 3 min and 134 min, respectively, of duration ranging 2sec–9 min each. Moreover, we observed that these honk samples were mostly collected in a controlled environment, free from ambient noise. These models were reported to produce classification accuracy of (73–97)%. However, some of the CNN-based deep learning models like SB-CNN [12], CNN [16], TFCNN [18], when trained using noise-free UrbanSound8K honk samples and tested using noisy honk samples collected by us produce detection accuracy even <75% (refer Section 6.3, Table 10). Therefore, we can infer that neither FFT/MFCC nor existing deep learning models are well suited for detecting honk in the presence of ambient noise.

Issues & Challenges: The broad system level challenges which we face to improve the honk detection accuracy in the presence of ambient noise are two folds. ***First, preparation of suitable training dataset***. The existing honk sample dataset (ESC-50, UrbanSound8k [19]) is inadequate, and that too mostly recorded in the absence of noise. Manually tagging honk samples from the raw traffic audio data are error-prone and cumbersome. Moreover, collecting raw traffic audio data using crowdsourcing to ensure enough spatio-temporal and vehicle type variety is also an important issue. ***Secondly, choice of suitable models for honk detection***. Honks originating from different vehicular modes exhibit different intensity, pitch, duration, loudness, etc. Therefore, determining suitable threshold values that cover all such categories is nontrivial. Ambient noise like extraneous speech, loud music, bioacoustics noise, electrical noise, etc., might partially mask the honk signatures in the recorded signals, which might lead to difficulty in distinguishing or filtering the honks from raw audio

samples. Hence, it is important to tune existing machine/deep learning models and prepare a suitable honk dataset of sufficient volume which can be used to train such models.

Contribution: This paper addresses the above challenges and presents a novel framework De$\mathcal{H}$onk, which efficiently detects vehicular honks in presence of road traffic. We have made contribution in both dataset preparation and also in tuning a suitable transfer learning model (ResNet152) that outperforms existing models. The specific contributions are highlighted as follows:

- We have collected raw audio samples using two volunteers via our customized android application AudREC covering 461 km spanning ∼13 km road segment in a suburban city of Durgapur, India, ensuring sufficient spatio-temporal variety. We created 7116 samples manually labeled by two personnel (refer Section 4).
- Noting this labeled sample size to be inadequate to train any deep learning models, we used this sample set to train a semi-supervised generative adversarial networks (SGAN) model [20,21], which subsequently labeled the remaining ∼16 hours of data, creating a collection of ∼56,000 labeled samples. Further, we used suitable time/frequency mask augmentation technique [12] to enhance the sample set to ∼1,80,000. The rich curated dataset (compared to 8000 labeled samples in existing UrbanSound8k) of significant volume thus created is used in our experiments and can also be a vital resource for other researchers (refer Sections 5.2 and 5.3.2).
- Inspired from the literature review, which reports the efficacy of spectrograms over MFCC in sound classification, we studied the usefulness of several transfer learning models namely VGG16 [22], VGG19 [23], MobileNet [24], ResNet50 [25], ResNet152 [26], Inception V3 [27] (refer Section 5.5). We found that ResNet152 with 1 dense layer to be the best performing model achieving honk detection accuracy 97.68% even in presence of the ambient noise (refer Section 6.1).
- Performance comparison with existing traditional CNN-based models SB-CNN [12], CNN [16], and TFCNN [18] reveal that transfer learning, in general, may be a better approach for detecting honks with >(7%–21%) (refer Section 6.2).
- We have shown that honk features detected using the proposed ResNet152 model can be used to successfully reveal interesting outdoor contextual features like less traffic, high traffic, marketplace, etc. Such information can be used to design effective novel micro-services like honk-aware route recommendations (refer Section 7).

The rest of the paper is organized as follows: in Section 2, some essential studies in the literature have been presented. An entire framework of the proposed work is provided in Section 3. Data collection, characteristics of audio data, Spatio-temporal characteristics of honks are discussed in Section 4. In Section 5, we provide an outline of spectrogram generation, data preprocessing, feature selection, and model training. Section 6 presents the outcome of each trained model along with the ground truth verification of the result. Implementation of honk-aware route recommendation is described in Section 7. Finally, in Section 8 we conclude the paper and discuss some future aspects of our work.

2. Related work

Many studies on noise/sound sensing and exposure level estimation have been conducted over the past few years by different researchers. We summarize them into three categories. In Sections 2.1 and 2.2, we discuss smartphone-based and infrastructure-based noise pollution sensing techniques respectively, whereas, Section 2.3 highlights the existing honks identification methods.

2.1. Smartphone-based noise pollution sensing

A number of works have been proposed for monitoring and measuring noise pollution exposure. Smartphone-based sensing is most popular among them due to its handy and mobility features. In [28], the author discussed a methodology to measure the participatory noise data using mobile phones, whereas in [29], smartphone-based noise level monitoring is discussed, and effective urban planning based on the corresponding noise pollution exposure is highlighted. In another work, [30], ambient noise monitoring using smartphones is addressed where the users usually roam around different locations at different times for collecting the data. Authors, in [6,31], focused on spatio-temporal correlation and determined the dynamic noise level pollution from the found correlation. However, smartphone based sensing is handy; sometimes, it may create problems when users move. Moreover, the sensitivity of the sensors in mobile phones may vary, and sensing information depends on the mobility pattern. Due to the GPS error, we can also lose some spatial information.

2.2. Infrastructure-based noise pollution sensing

Apart from smartphone-based noise level monitoring, fixed sensor-based pollution level monitoring methods are used in many works due to their reliability and efficiency. In these techniques, the sensors are placed in different locations of the road segments to measure the noise data. In [32], noise pollution is monitored and analyzed in an urban area using fixed sensors. However, in [33], traffic prediction in road traffic areas is proposed by exploiting the noise pollution levels. A sound and air pollution monitoring system with temperature and humidity sensing and raspberry Pi controlled cloud-based sensor is investigated [34]. Arduino controller with IoT technology can be used as a fixed sensing technique to monitor noise data, and it is explored in [35]. In some recent works, wireless sensor-based methods have been used

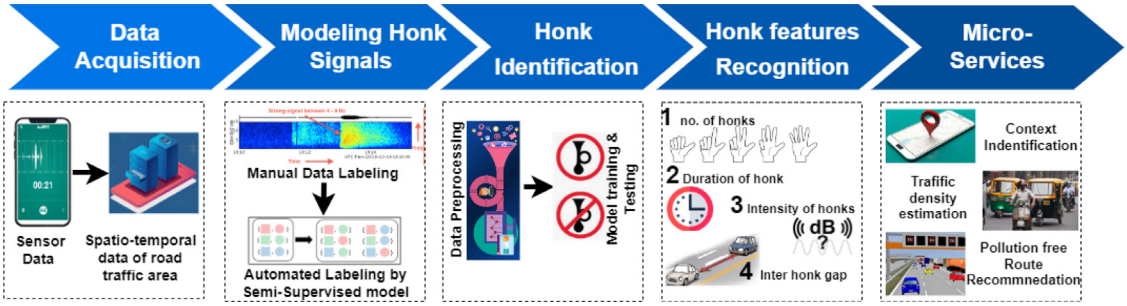


Fig. 1. Framework for our proposed system.

to measure noise pollution levels. In [36], the authors designed a wireless sensor network (WSN) based application to determine the noise pollution levels. A similar kind of work is also found in [37]. A survey work on noise pollution monitoring using sensor networks is presented in [38]. In [39], the authors identified the sound source and classified them to determine the environmental noise pollution level. The techniques stated above have both advantages and drawbacks. The main advantage is that it gives us a temporal noise pattern. The drawbacks include the additional cost of deploying & maintaining the sensors, low spatial coverage, and the sensors may be damaged or stop working due to several environmental hazards like storms, rainfall, etc.

2.3. Vehicular honk detection

Many research works highlight that vehicular honks are the major reason behind excessive noise pollution in urban areas. Hence, honk identification and honking feature characterization are becoming essential for developing various applications to mitigate noise exposure levels. Works related to honk identification or recognition can be divided into two categories: (1) environmental sound classification, where vehicular honk is one of the classes, and (2) algorithms to recognize honks directly from the audio samples. Several works [11–18] related to environmental sound classification using deep learning models are summarized in Table 2. Works in [14], and [16] performed spectrogram-based sound classification, where, in others, audio samples were converted into either log-scaled mel-spectrograms or MFCC. Most works applied CNN model and its variations for classifying sounds in which the car honk is treated as a class. On the other hand, a limited number of works have been done to identify vehicular honks without applying deep learning or machine learning techniques. For example. in Horn-Ok-Please [8], the authors used FFT to identify the number of honks and determined vehicular speed based on the honks. They used band-pass filters for removing ambient noise from the noise samples where the lower cut is 2 kHz, and the uppercut is 4 kHz. However, it does not guarantee that all frequencies in this range would signify a honk. Moreover, they did not propose any framework which can help to identify the honks automatically while moving. A similar kind of technique is discussed in [10,40] to determine car honks based on a smartphone based application to assist hearing-impaired people. In another work, MFCCs based technique [9] is used for honk identification. Apart from FFT and MFCCs, an embedded system based emergency honk detection is developed in [41], where the authors proposed an algorithm embedded into the myRIO platform using its FPGA. The qualitative analysis of the above cited [8–10,40,41] techniques is presented in Table 2. Apart from these studies, as microservices, noise pollution free route recommendations were also developed by the researchers in [6,42] where noise pollution was measured based on sound pressure levels.

From the above review of the work, we observe that 2.1 and 2.2 mainly focus on noise pollution sensing and modeling, and Section 2.3 gives honk identification using FFT, MFCC and different deep learning models. However, it is apparently clear that no such works have developed a framework so far, which can identify vehicular honks automatically from a raw audio sample in the presence of ambient noise. Moreover, extracting contextual information and developing micro-service based on different honking features is not yet considered till yet.

3. Framework

The proposed framework De$\mathcal{H}$onk contains four different modules, as depicted in Fig. 1.

- **Data collection**: In our work, data acquisition plays an indispensable role in carrying out our tasks. A customized android based application is developed for measuring the honks data from the environmental sources. Further, spatio-temporal characterization of honk data has been done to analyze the inter-dependency among the honk features.
- **Modeling**: From the raw sample audio data, manual data labeling is done, spectrograms are generated, and afterward pre-processing is performed, resulting in training and testing set. Then, these spectrogram images are fed into different deep learning models for training. Based on the statistical inference best-suited model is selected.

Table 2
Literature review for environmental sound classification and vehicular honk detection.

Author, Year	Dataset	Technique	ML/DL	Outcome	Remarks
Sound classification					
Piczak et al. (2015) [11]	ESC-50, ESC-10, and UrbanSound8K	Log-scaled mel-spectrograms	CNN (hidden layers as ReLUs and a softmax is used in output layer.)	10 different class of environmental sound is classified	Car honk included
Salamon et al. (2017) [12]	UrbanSound8K	Log-scaled mel-spectrograms	SB-CNN (modified version of [11])	10 different class of environmental sound is classified	Car honk included
Zhou et al. (2017) [13]	UrbanSound8K	2D mel-spectrogram	ConvNet a CNN model (modified version of [12], which can learn temporal-spectral features)	10 different class of environmental sound is classified	Car honk included
Khamparia et al. (2019) [14]	ESC-10 and ESC-50	Spectrogram	CNN and tensor deep stacking network (TDSN)	10 different class of environmental sound is classified	Car honk not included
Mesaros et al. (2019) [15]	freesound.org, TUT Sound, and AudioSet	log-mel energies, MFCC, pitch, spectrogram	CNN, DNN, RNN, CRNN, LSTM,GRU	Classified 4 categories of sound in three different phase	Car honk included
Demir et al. (2020) [16]	DCASE-2017 ASC and UrbanSound8K	Spectrogram	Transfer learning models	For 1st dataset 15 category and 2nd dataset 10 category classification is done	Car honk included
Ahmed et al. (2020) [17]	ESC-50, ESC-10, and UrbanSound8K	Log-scaled mel-spectrograms	CNN	10 different class of environmental sound is classified	Car honk included
Wenjie et al. (2021) [18]	ESC-50, and UrbanSound8K	Harmonic spectrogram, Percussive spectrogram	TFCNN	10 different class of environmental sound is classified	Car honk included
Honk detection					
R. Sen et al. (2010) [8]	Self collected	FFT	Self design honk detection algorithm	Vehicular honk identified	No ML/DL is used
R. Banerjee et al. (2012) [9]	Self collected	MFCC	Modified MFCC	Vehicular honk identified	No ML/DL is used
Takeuchi et al. (2014) [40]	–	IIR Comb Filter	Self design honk detection method	Vehicular honk identified	No ML/DL is used
Josef et al. (2016) [41]	–	Embedded system	Self design honk detection method	FPGS Based emergency honk identification	No ML/DL is used
Dim et al. (2020) [10]	–	FFT	Self design honk detection algorithm	Vehicular honk identified	No ML/DL is used

- **Result analysis and inference**: In this module, the accuracy of different models is tested and based on the most affirmative outcome of the model, the ground truth of our work is verified with a physical location.
- **Micro-services**: Depending upon the number of honks, duration of honks, inter honk duration, and sound pressure level, the context of a location can be inferred. In addition, a micro-service named honk-aware route recommendation is developed to ameliorate living standards.

4. Data collection

Honking behavior, pattern, and number of honks change over time and space. Hence, widespread data collection based on spatio-temporal characteristics is important to develop a honk profile and micro-service. In the proposed work, we collected noise data using our customized android application, where we set the sampling rate to 8 kHz, bit depth to 16bits, and used audio channel as mono, and audio format as Wave(.wav). During data collection, the sounds that we met in the urban environment and appeared in the audio files were engine noise, human voice, vehicle honk, background music, and other traffic noise. Moreover, we collected our data from various places, which include high and low traffic areas, marketplaces, shopping malls, hospital locality, bus terminus, educational institutes, and highways in Durgapur,[1] a

[1] https://en.wikipedia.org/wiki/Durgapur.

Table 3
Represents the details information of data collection routes.

Route	Duration of audio data	Distance covered per trail	Mode of transport	Total distance covered	Demographic profile
Route1	736 min	22.8 km	Bus	421.8 km	Less crowded and high crowed traffic, Highway, Market places, Bus terminal, Educational institute, Hospital area
Route2	223 min	3.5 km	Cycle	39.5 km	Market places, Less crowded and high crowed traffic, Traffic junction

Table 4
Traffic profile of the data collection routes.

	High traffic zone								
	Highway			Market place			Low traffic area		
	Morning	Afternoon	Evening	Morning	Afternoon	Evening	Morning	Afternoon	Evening
Traffic volume (Vehicle/h)	270–330	220–280	270–330	160–180	100–120	110–220	100–140	100–130	100–140
Average Volume/day	3000			2000			1200		
Number of car lanes	2–4			2			2		
Type of vehicles	Bus, Truck, Motorcycle, Car			Bicycle, Motorcycle, Car, Autorickshaw			Bus, Motorcycle, Bicycle, Three Wheeler, Autorickshaw		
Type of area	Bus terminals, Highway			Different market area			Near to residential place		

sub-urban city in India. We reached 3.5 km to 22.8 km road segment in every slot, and at least 461 km (approx.) distance was covered in total. To assess the honking characteristics, each day is subdivided into three time slots (i.e., morning, afternoon, and evening). Furthermore, we took different modes of transport to collect samples, such as bus to cover the longer route and cycling and walking for the shorter route. A total of 16 h of data have been collected. Table 3 represents the demographic characteristics of the route and the detailed data collection information, whereas Table 4 describes the characteristics of high traffic and low traffic area in terms of traffic volume, number of car lanes, types of vehicles etc.

4.1. Spatio-temporal characteristics of honks

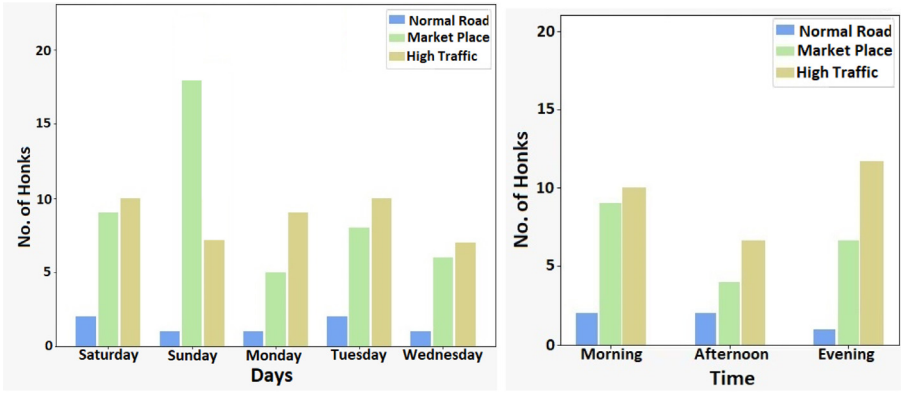
In this section, we have explored some intuitive honk features and studied their spatio-temporal characteristics from the collected data samples. We have considered three different places (less traffic, marketplace, high traffic) and three time slots (morning, afternoon, evening) to identify various honk features such as number of honks, duration of honk, inter honk gap, and sound pressure level. We monitored the data for 5 min in each time slot and performed manual feature tagging to understand the existing correlation between them.

4.1.1. Honk count estimation

Measuring the number of honks at a particular location is an essential parameter in designing a honking profile. A sample case study is taken into account where we have considered three different places such as irregular traffic movement, marketplace, high traffic road, and the results are plotted in Figs. 2(a) and 2(b). Fig. 2(a) represents the temporal plot that captures the changes in the honks count over a fixed time and location on different days of a week. Here, data samples were collected between 9:00 am to 9:30 am for 5-consecutive days starting from Saturday to Wednesday, and we framed 2-min data into the plot for all the locations. The figure reveals that the number of honks is almost the same for all the days, but in marketplace areas, it goes high on the weekend. In contrast to the marketplace, a comparatively lower number of honks are noted in high traffic areas on the weekend. On the other hand, Fig. 2(b) depicts the spatial plot that shows the variation of the honks count during different time slots of the same day at the exact location. Data were monitored from 9:00 am to 9:30 am in the morning, 1:00 pm to 1:30 pm in the afternoon, and 6:00 pm to 6:30 pm in the evening, and we plotted 2-min data for each slot. At the normal traffic, the same output is observed as shown in Fig. 2(a). However, the number of honks experienced at the marketplace and high traffic areas is much higher during the morning and evening as compared to the afternoon.

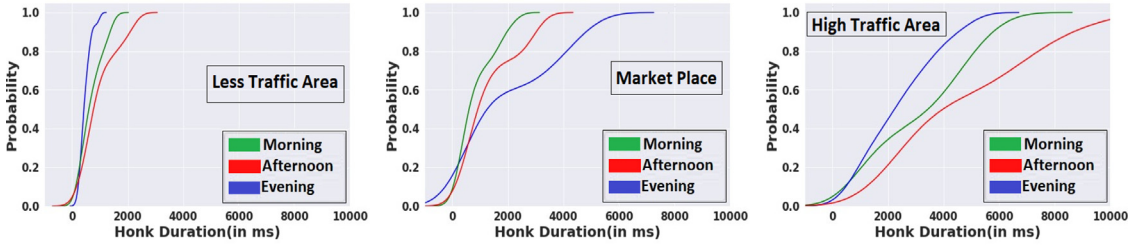
4.1.2. Distribution of honks duration

The kernel density estimation (KDE) plot is used to visualize the distribution of honk duration during the morning, afternoon, and evening. Fig. 3(a) represents a normal traffic area where distribution mainly lies between less than 1 sec



(a) (b)

Fig. 2. (a) A plot of honk sample in different days during the same time 9 am to 9:05 am, (b) A plot of honk sample in three different time slot throughout a day.



(a) (b) (c)

Fig. 3. Distribution of honk duration of (a) Less traffic area (b) Marketplace, and (c) High traffic area for three different time slots.

to 2.5-sec duration, and we see that the honk duration is almost the same for the entire day. Fig. 3(b) shows a marketplace that contains 1 sec to 7-sec span. The distribution of honk duration in this area has reached a maximum in the evening. Fig. 3(c) depicts a high traffic area that includes 2 sec to 10-sec duration of honks, and the honk duration is high throughout the day. From the figures, we have observed an interesting feature, i.e., the duration of honks varies throughout the day in all the places, which gives a strong correlation.

4.1.3. Inter honk duration

Inter honk duration between two successive honks is another important feature in order to identify the context of a region since a long gap between the honks usually refers to low traffic areas. In 2-min time slice, we have observed that gap between the honks lies between 1 sec to 20 sec window, and this window is further divided into 4 buckets where the first bucket contains 0–5 sec honk duration, the second bucket contains 6–10 sec honk duration, and so on. The first bucket represents a small gap between the honks, containing a maximum number of high traffic data points, whereas none of the high traffic data points is in the last bucket because a longer gap between successive honks in high traffic does not usually occur. Fig. 4 represents the detailed observation of three different areas to understand variation in inter honk duration.

4.1.4. Sound pressure level

Sound pressure level (SPL), an important feature of honking noise, is used to assess the environmental pollution level. SPL is measured in decibel (dB) value. We have monitored the SPL for three different time slots in a high traffic area. Fig. 5 shows the variation of SPL of a day. It is observed from the figure that, for the morning, the span of SPL is 70 dB to 100 dB, but the SPL is mainly concentrated on 80 dB to 90 dB. On the other hand, during afternoon and evening, SPL ranges from 65 dB to 85 dB and 75 dB to 90 dB, respectively. These temporal variations of SPL illustrate its importance in characterizing pollution levels.

4.1.5. Correlation between sound pressure level with other honk features

The Pearson correlation coefficient (PC) is used to analyze the correlation with SPL and all other honk features for morning slot of high traffic areas, as depicted in Fig. 6. Fig. 6(a) shows the moderate positive correlation (PC = 0.57)

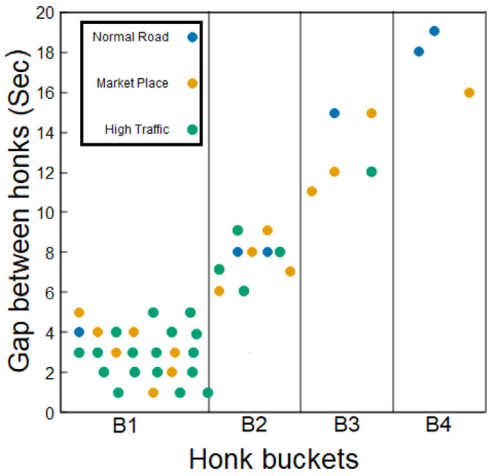


Fig. 4. Successive difference between honks in less crowed Road, Market place, and High traffic area.

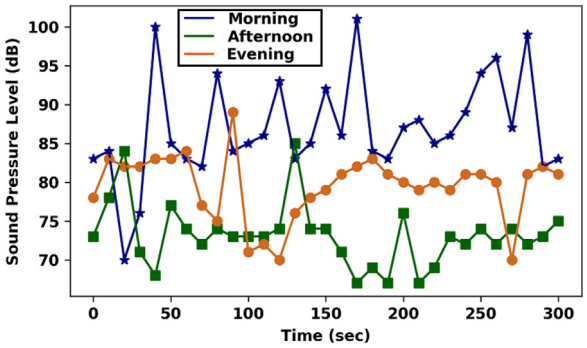


Fig. 5. SPL of a day in High traffic area.

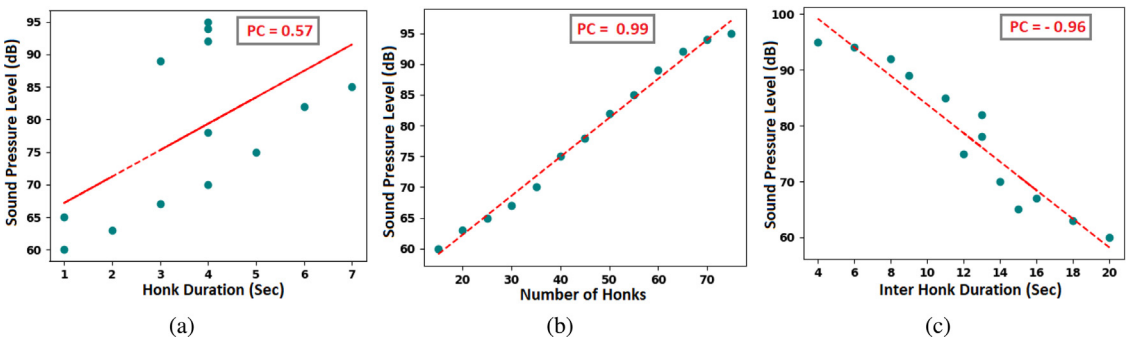


Fig. 6. Correlation between SPL with (a) Honk duration (b) Number of honk, and (c) Inter honk duration.

between the duration of honks and SPL that signifying high duration does not refer to the high SPL all the time, whereas a very strong positive correlation (PC = 0.99) is found in between the number of honks and SPL in Fig. 6(b). In contrast to these plots, Fig. 6(c) represents a very strong negative correlation (PC = −0.96) between inter honk duration and SPL.

The experiment above illustrates a good correlation among all the honk features in multiple circumstances based on their spatio-temporal characteristics. However, manual analysis of such characteristics is difficult. Hence, we need to develop a system that automatically extracts these features from raw noise samples.

5. Methodology

This section describes the method of honk identification and its feature selection. At first, we introduce the modeling of the honk signal, followed by data labeling and preprocessing. After that, we discuss various deep learning models to identify honks and determine an appropriate model for our proposed method.

5.1. Modeling the honk signal

To visualize the audio data in a succinct form of an image, we have generated a spectrogram by splitting each audio sample into a 1-sec segment, where the x-axis indicates time, the y-axis represents frequency in Hz, and the different colors are used to represent the amplitude of each frequency. Here, the light color signifies a high amplitude compared to the dark color. In our work, we have converted all the audio data files into spectrograms and fed them into deep learning models for further processing. For example, a sample 10-sec audio clip data is converted into a spectrogram and shown in Fig. 8, where four possible honks are reflected in the image, and the frequency of the honks lies in the range between 2–4 kHz.

Many works have shown the advantages of using spectrograms and deep learning models in different applications such as emotion recognition from speech [43], multi-class language identification from the audio signal [44], sound classification based techniques [14], even early COVID-19 detection [45]. However, the use of spectrograms and deep learning models in vehicular honk identification is not yet studied, which might be an interesting solution in our case, and we have addressed this in this paper.

5.2. Label data preparation

One of the crucial tasks in our work was data labeling, as raw audio data contains various ambient noises, and sometimes different vehicular honks might be overlapped in many situations. Due to the unavailability of labeled data, we initially performed manual data labeling. As manual labeling is usually time consuming and error-prone, we employed two personnel separately to perform the task. In the end, we verified both files to minimize unintentional errors. Here, all audio files were divided into 1-sec clips, and by listening each clip, we marked the honk as the positive class for the classification and assigned the label '1'. All other noises were grouped together into a single class for classification and assigned the label '0'. Sometimes it might be challenging to rectify the presence of honk when the amplitude of honk is less than the ambient noise. In that case, if one personnel marks a honk as '1' and the other marks as '0', then we consider '1' for the particular sample tagging. Personnel 1 identified 815 honk samples, whereas Personnel 2 detected 825 honk samples. A total of 7116 samples are labeled manually.

Data Labeling using SGAN: Manual data labeling of 16 h of data is a challenging, error-prone, and time-consuming task. For that reason, we have employed a semi-supervised learning framework (i.e., Semi-supervised generative adversarial network (SGAN)) to label the noisy samples. In semi-supervised learning, a small amount of labeled data is used to train a model from a vast volume of unlabeled samples, such that it can learn using both labeled and unlabeled samples. In our work, we have used manual labeled data as an input for SGAN. Like generative adversarial networks (GANs), SGAN also uses two neural networks i.e., generator and discriminator. These two are used to complement each other. The generator intends to generate real and fake spectrogram-like images to make the discriminator commits a mistake. The model consists input layer followed by a dense layer and a LeakyReLU layer to give a basis for 16×16 image. The following three convolutional transpose layers unsample the initial image to obtain a 224×224 image just like the images in the dataset. The output is given through a convolution layer with a tanh activation function. The discriminator of SGAN has two parts: the supervised classifier, which will perform honk classification, and the unsupervised one is used to identify the real and fake images. The model consists of an input layer followed by three convolution layers, each followed by a LeakyReLU layer. Finally, they each have a dense layer containing only one node since our task is of binary classification. These dense layers each have a sigmoid function. The detailed architecture of the SGAN model is shown in Fig. 7. Using SGAN, we have labeled 56750 samples.

5.3. Preprocessing

5.3.1. Screening of spectrogram images

A spectrogram is obtained for each small segment using a raised cosine window of length 256 samples (32 ms at 8 kHz) and hop length 64 samples. Output is a complex matrix that is transposed to obtain time-wise frequency amplitudes. In a conventional way, the absolute value of the amplitudes is log-scaled to improve visualization and feature interpretation. The resultant matrix is scaled using Min–Max Scaling so that values fit in an 8-bit range used to denote pixel intensity values in images. Values are also inverted by subtracting from 255 so that the dark areas in the image represent regions of higher energy for the corresponding frequency components. Each spectrogram is exported as a Portable Network Graphics (PNG) file. The process is repeated for each segment of each audio clip.

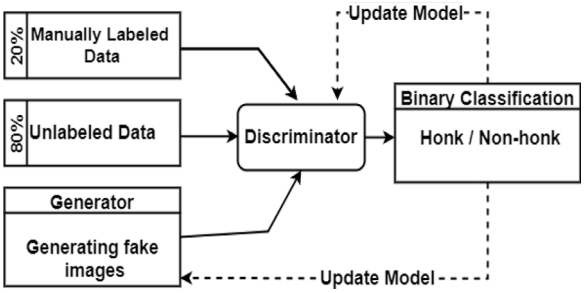


Fig. 7. Illustration of SGAN model that converts unlabeled data to labeled data and identifies the honks from it.

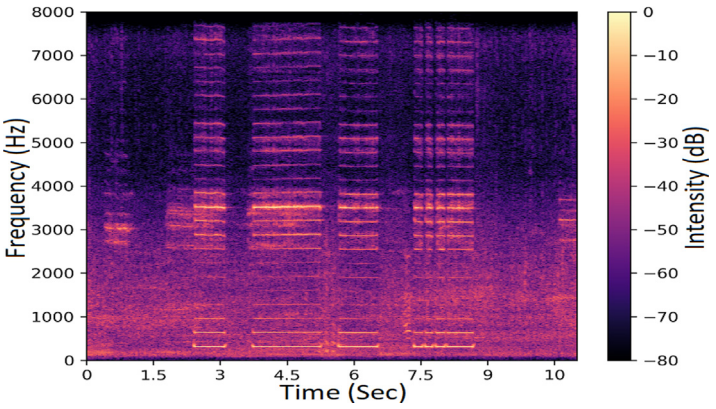


Fig. 8. Sample spectrogram of a road traffic area, that represents four honks in absence of ambient noise in the presence of ambient noise for 10 s duration.

Table 5
The amount of data and its distribution for manual data labeling, automated data labeling, data balancing and the data for model training and testing.

Class	Manually label data	Labeled by SGAN	After augmentation	After balancing	Distribution of data for training and testing set	
	No of samples	No of samples	No of samples	No of samples	Count in training set	Count in testing set
Total	7116	56750	180738	77238	61789	15449
Non-honk	6291	47373	142119	38619	30893	7723
Honk	825	9377	38619	38619	30896	7726

5.3.2. Data augmentation and class balancing

In our work, we faced an imbalanced classification problem due to the difference in sizes of two classes, i.e., honks and non-honks. One of the ways to deal with this problem is dataset balancing. Out of 63866 total samples (combining manual and SGAN labeling), the dataset contains only 10202 samples (i.e., 16% of the entire dataset) that carry honk signature. If we want to balance the dataset, we can lose a vast amount of data. Hence, we performed data augmentation techniques to increase the honk signature samples in the dataset artificially. Data augmentation improves data diversification. In this research, we have used time mask and frequency mask as augmentation techniques. In case of a time mask, some vertical bars are placed randomly to conceal the ranges of time in the spectrogram image. A similar approach is performed in frequency mask, where we have added horizontal bars randomly to mask a range of successive frequencies in the spectrogram. Fig. 9(a) shows the time mask augmentation on Fig. 8 spectrogram image, where the yellow horizontal bar masks a time slot of the spectrogram. The frequency mask feature is shown in Fig. 9(b), where the yellow vertical bars hide a frequency signature from the spectrogram image.

After augmentation, the number of honk samples increased from 9377 to 38619. We randomly selected the same amount of non-honk samples for class balancing. A total of 21 h of data has been considered for model training and testing. Finally, we performed the train-test split of 33% to create the test dataset that would be used to evaluate the performance of the model on unseen data. The distribution of data is shown in Table 5 in detail.

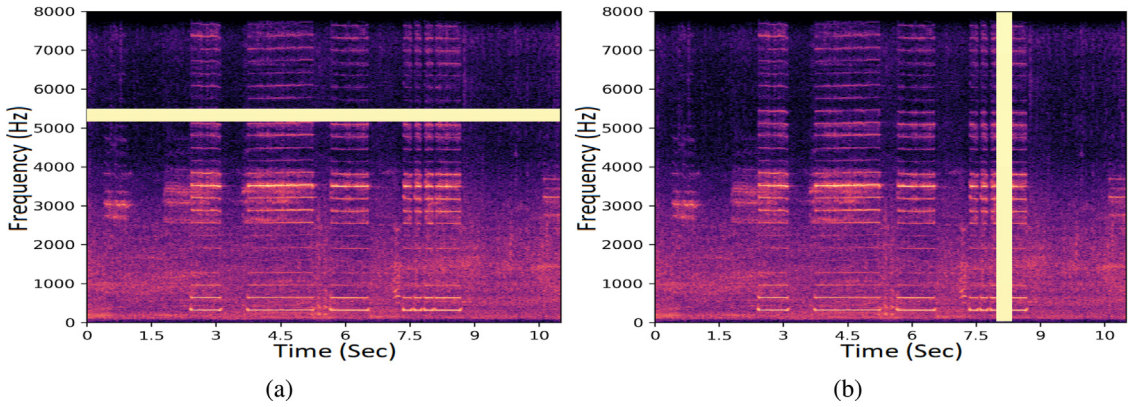


Fig. 9. Spectrogram are generated by applying (a) Time mask and (b) Frequency mask of data augmentation techniques.

5.4. Feature selection

Feature selection is an essential step in our work. The followings are the honk features used in this paper:

$$Fh_L \in \mathcal{T} \wedge \mathcal{IH}\ [\mathcal{N}(h_i),\ \mathcal{L}(h_i),\ \mathcal{G}(h_i),\ \mathcal{S}(h_i)]$$

where Fh_L is the feature of honks for a fixed location, which is belongs to three parameter of honk information ($\mathcal{IH}$) and the time ($\mathcal{T}$).
$\mathcal{N}(h_i) = i^{th}$ number of honk;
$\mathcal{L}(h_i) =$ length of ith honk;
$\mathcal{G}(h_i) = \mathcal{T}(h_{(i+1)} - h_i)$ i.e. time gap of two successive honks;
$\mathcal{S}(h_i) = SPL\ of\ i^{th}$ honk;
The features value are automatically changed if the time or space is varied.

5.5. Choice of model for honk identification

Deep learning utilizes multiple-layer architectures to extract inherent features in data, and they can discover massive amounts of structure in the data. Deep learning in image analysis, processing, and new image recognition has been widely discussed in many studies [23–25]. CNNs are used for this purpose. Moreover, in some recent works, deep learning models are introduced to analyze the spectrogram to detect COVID 19 [45]. Another approach is to use Transfer Learning, where models are pre-trained with ImageNet dataset [46], an extensive dataset of more than 14 million image data. Although the researchers analyzed the spectrogram images with deep learning models in their research, the vehicular honks identification using the same has not yet been well studied. Hence, we aim to investigate *if the transfer learning models be used effectively to detect the honk more accurately even in a noisy environment?*

We have selected four popular pre-trained models like VGG, MobileNet, Inception V3 and ResNet to detect the vehicular honks from audio spectrograms. The selected pre-trained models are available as part of the trusted public libraries such as Keras and pre-trained with the ImageNet dataset, a large dataset of more than 14 million image data. Further, these models are classified into two groups, i.e., lightweight models and heavyweight models based on the number of convolution layers. The models like VGG16, VGG19, and MobileNet are considered as lightweight models, while, as heavyweight models, we have used ResNet50, ResNet152, and Inception V3. It is worth mentioning that lightweight models have fewer convolution layers that reduce the complexity and provide faster computation. In contrast, the heavyweight models consist of more convolution layers for better accuracy. The model details used in the proposed work are given below:

5.5.1. VGG Nets (VGG16 & VGG19)

The VGG16 and VGG19 models have smaller network architecture, which is usually more beneficial for a domain-specific dataset. VGG16 model has 13 convolution layers and 3 dense layers, whereas VGG19 has 16 convolution layers followed by 3 dense layers.

5.5.2. MobileNet

MobileNet is a convolution neural network architecture, usually considered to design lightweight deep convolutional neural networks, which are resource-friendly and can run on mobile phones and embedded applications. It uses depth-wise separable convolution filters composed of point convolution filters and depth-wise convolution filters. In each input channel, the depth-wise convolution filter conducts a single convolution, where the point convolution filter uses $1 * 1$ convolutions linearly by combining the output of depthwise convolution filters.

Table 6
Performance analysis of lightweight models (VGG16, VGG19, and MobileNet models).

Evaluation metric	VGG16	VGG19	MobileNet
Accuracy	92.51%	91.04%	91.48%
F1 Socre	0.93	0.92	0.91
F2 Score	0.95	0.95	0.88
MCC	0.85	0.83	0.83
ROC AUC	0.93	0.91	0.91
Precision	0.89	0.86	0.97
Recall	0.96	0.98	0.86
Kappa statistics	0.85	0.82	0.83

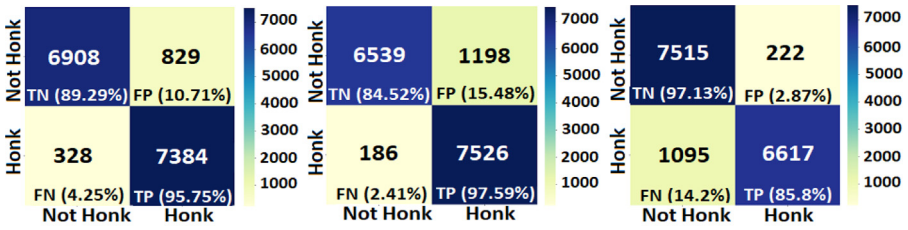


Fig. 10. Confusion matrix of (a) VGG16 (b) VGG19 (c) MobileNet.

5.5.3. ResNets (ResNet50 & ResNet152)

ResNets (Residual Networks) have made a massive breakthrough in image analysis. It uses more convolution layers to solve more complicated cases by removing the vanishing gradient problem from deep neural networks. Residual learning (skipping the connections between layers) is used to solve the vanishing gradient problem. ResNet50 implements 50 layers of residual learning, whereas ResNet152 uses 152 layers of residual learning.

5.5.4. Inception V3

The Inception V3 is 48 layers deep neural network that improves the utilization of computer resources by expanding the depth and width of the network without changing the computation operations. It uses the concept of an inception module that optimizes the network structure by skipping the connection. To reduce the dimensional, the inception module is repeated by stacking with max-pooling layers.

For all the pre-trained models, we have used only one dense layer to avoid the problem of overfitting.

6. Result analysis

We have evaluated the performance of the proposed technique in Section 6.1 using deep learning models (transfer learning) discussed in Section 5.5 through a series of experiments. We have used four performance indicators i.e., F1 score, F2 score, Matthews correlation coefficient (MCC), Area under ROC curve (ROC AUC), Precision, Recall, and Kappa Statistics to assess the models. Comparison with the baselines is discussed in Section 6.2. In Section 6.3, the performance of the selected transfer learning model is analyzed with a standard dataset. Smartphone sensitivity for detecting honks is also discussed in Section 6.4. In addition, a glimpse of the probable location-based context is emanated based on the characteristics of honk data, and a ground-truth verification is also verified with a physical location (Section 6.5). In Section 6.6, we have summarized the importance of our work.

6.1. Performance analysis of models

In order to evaluate the performance of all the models, we have used 77 238 labeled sample data, where 38 619 non-honk samples and 38 619 honk samples are used for training and testing purposes. As an experimental setup, we used 224 $\times$ 224 $\times$ 3 image size, 20 epochs for all models, and batch size 312 for all the models. We first evaluate the performance of the lightweight models and then we execute heavyweight models in honk identification. The evaluation metrics for all the lightweight model are shown in Table 6. We see that the accuracy achieved by VGG16, VGG19, and MobileNet are 92.51%, 91.04%, 91.48%, respectively, and MobileNet exhibits poor performance among all. It is also noticed that VGG16 slightly outperforms than VGG19 and MobileNet in terms of all evaluation metrics. In Fig. 10, we have shown the confusion matrix for analyzing the performance of a classification model on a set of test data for known true values, where VGG19 has the highest True-Positive (TP) rate (97.59%) but the lowest True-Negative (TN) rate (84.52%). Overall, the TP and TN rate of the VGG16 model is more acceptable when it is compared to other two models.

Table 7
Performance analysis of computationally intensive models (ResNet50, ResNet152, and Inception V3 models).

Evaluation metric	ResNet50	ResNet152	Inception V3
Accuracy	96.66%	97.69%	83.58%
F1 socre	0.97	0.98	0.85
F2 score	0.96	0.98	0.89
MCC	0.93	0.95	0.68
ROC AUC	0.97	0.98	0.84
Precision	0.97	0.97	0.79
Recall	0.96	0.98	0.92
Kappa statistics	0.93	0.95	0.67

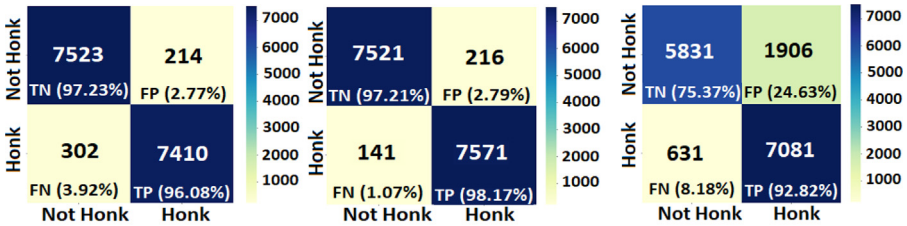


Fig. 11. Confusion matrix of (a) ResNet50 (b) ResNet152 (c) Inception V3.

To improve the accuracy further in honk identification, we have tested the heavyweight models such as ResNet50, ResNet152, and Inception V3 on our dataset. Table 7 tabulates the experimental outcomes of each model, where we observe that RestNet152 provides higher accuracy (97.69%) compared to RestNet50 and Inception V3. In Fig. 11, we provide the confusion matrix generated by the heavyweight models on our dataset for further analysis. In ResNet50 and ResNet152, the TN rate (97.2%) and FP rate (2.7%) are the same. However, TP rate is increased by 2%, and the FN rate is decreased by 2% in case of ResNet152. This result shows that in a few cases, some ambient noises are recognized as honks and the honks are misclassified in negligible cases. Epoch-wise training accuracy obtained by each model is depicted in Fig. 12 where the accuracy increases with each number of epochs, and it is clearly observed that ResNet152 performs better than other models. From Tables 6, 7 and Fig. 12, we can say that the RestNet152 model has obtained good results in identifying vehicular honk from the raw audio samples, and hence, from now onwards, we will focus and restrict the rest of the experiments on the ResNet152 model only.

Besides this, we evaluated all pre-trained models and checked the accuracy where the models were not trained with our dataset but tested on our dataset. Fig. 13 shows the accuracy of the models with and without training on our dataset. It is noticed that the accuracy level of the pre-trained models drops below approx. 40% when they are not trained on our dataset. The accuracy falls sharply as the spectrogram images are only a fraction of the ImageNet dataset. Some previous studies considered traditional CNN variant models for environmental sound classification problems. Hence, the question is *can we use these models for detecting vehicular honks?* In the following section, we analyze the performance of ResNet152 and CNN-based methods in honk detection in detail.

6.2. Comparison with baseline work

We consider existing SB-CNN [12], CNN [16], and TFCNN [18] as baseline models to evaluate the performance of our proposed RestNet152 model with respect to various performance metrics. All these baseline models are CNN-based and use softmax as an activation function. Moreover, they are mainly used for multi-class classification. Since our work considers binary classification, we have extended the CNN model in [18] and named it CNN-BIN, and compared its performance with the proposed RestNet152 model. The CNN-BIN model contains seven convolutions layers between the input and output layer and two dense layers, where the first dense layer is used to reduce the feature and the second dense layer is for predicting honks and non-honks from the data. If the predicted value lies between [0 to 0.5] signifies non-honk and [0.6 to 1] signifies honk. Moreover, we have used sigmoid as an activation function and binary cross entropy as a loss function. The detailed configuration of all baseline models is shown in Table 8. Among all baselines, CNN [16] has better performance than the other three baselines on our dataset, as shown in Table 9. However, it is interesting to see that the ResNet152 model has achieved more than 7.71%, 10%, 10%, 20.25%, 10%, 6.59%, 10%, 20.25% improvement in terms of accuracy, F1 score, F2 score, MCC, ROC AUC score, Precision, Recall, and Kappa statistics respectively as compared to CNN model [16]. Overall, observations from the table show that the performance of RestNet152 is the best among the comparative models at detecting honks in the presence of ambient noise.

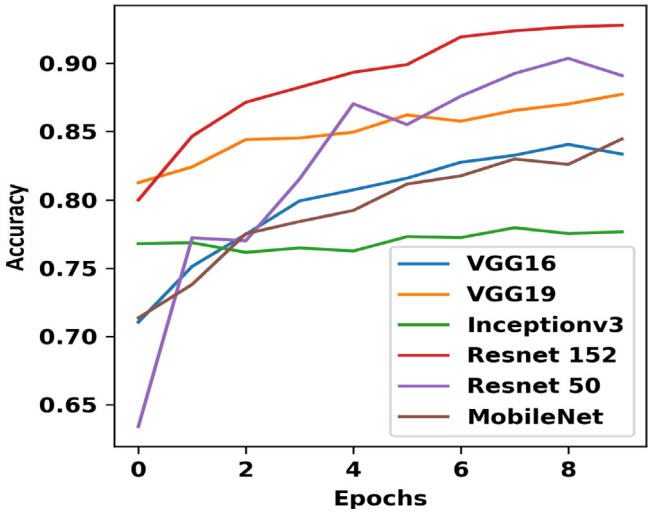


Fig. 12. Epoch vs. accuracy plot at the time of the training of each model.

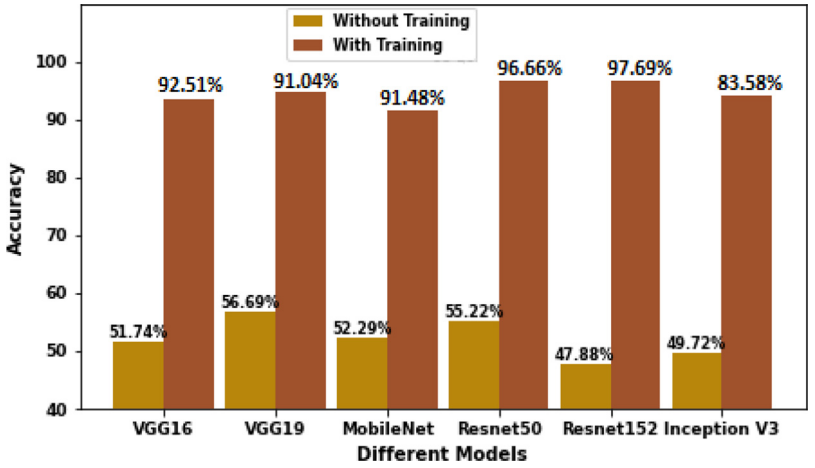


Fig. 13. Accuracy comparison of different models using pre-trained model with and without further training.

Table 8
Configuration of the baseline models.

Models	Conv2d	Max polling	Up Sampling	Dense Layer	Batch normalization	Activation function
SB-CNN [12]	3	2	–	3	3	Softmax
CNN [16]	6	3	–	2	–	Softmax
TFCNN [18]	3	2	–	2	–	Softmax
CNN-BIN	7	3	3	2	–	Sigmoid

6.3. Performance analysis of ResNet152 on standard dataset

We collected urban sound data from [19] as a standard dataset from where we extracted the vehicular honk related audio files. The longer file duration in [19] is 9 min 34 sec, whereas the duration of the smaller file is 13 sec In total 134 min of data were collected, which were further used to train the ResNet152 model. The obtained results are compared with our dataset and summarized in Table 10. It is seen that the accuracy level is almost same for both datasets. We have performed another experiment to illustrate the effectiveness of our data in successful honk identification. To do that, we trained the ResNet152 model with the urban dataset and then tested the model using our dataset. It is observed from Table 10 that the accuracy in honk detection is decreased by 11.69% since the urban dataset contains minimal ambient noise, which affects the accuracy of the trained model. A similar experiment is performed with baseline CNN [16], and the accuracy falls to 71.65%. This result shows that CNN [16] accuracy is not satisfactory in identifying vehicular honks when

Table 9
Comparison the performance ResNet152 model with baseline work.

Evaluation metric	Baseline work				Our proposed model
	SB-CNN [12]	CNN [16]	TFCNN [18]	CNN-BIN	
Accuracy	75.98%	89.98%	80.21%	80.21%	**97.69%**
F1 Score	0.77	0.89	0.8	0.79	**0.98**
F2 Score	0.78	0.89	0.78	0.78	**0.98**
MCC	0.52	0.79	0.61	0.61	**0.95**
ROC AUC	0.76	0.89	0.8	0.81	**0.98**
Precision	0.74	0.91	0.82	0.86	**0.97**
Recall	0.79	0.89	0.76	0.75	**0.98**
Kappa statistics	0.52	0.79	0.6	0.63	**0.95**

Table 10
Comparison analysis of our dataset with standard data source [19] in ResNet152 environment.

Evaluation metric	Training & Testing with our dataset	Training & Testing with external dataset	Training with external dataset & Testing with our dataset in ResNet152 & (CNN [16]) environment	Training with our dataset & Testing with external dataset by combining different environmental sound
Accuracy	97.69%	97.66%	86.29% & (71.65%)	96.85%
F1 Score	0.98	0.98	0.84 & (0.66)	0.97
F2 Score	0.98	0.96	0.76 & (0.60)	0.97
MCC	0.95	0.95	0.75 & (0.45)	0.94
ROC AUC	0.98	0.98	0.83 & (0.57)	0.97
Precision	0.97	0.97	0.88 & (0.81)	0.95
Recall	0.98	0.97	0.86 & (0.57)	0.98
Kappa statistics	0.93	0.95	0.77 & (0.43)	0.94

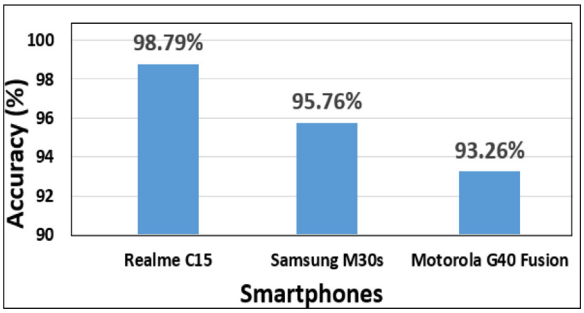


Fig. 14. Comparison of the accuracy among three different smartphones.

the input audio signal contains a lot of ambient noises. In addition, we have prepared another test dataset by combining honk data and other non-honk data (like air conditioner, drilling, dog bark, gunshot, children playing, etc.) taken from [19]. These non-honk data might resemble the signature of the honk. Therefore, a collective dataset is prepared by combining honk data with sound data, which resembles honk considering 953 honk samples and 953 other outdoor environmental samples. We have tested our trained model (i.e., ResNet152) using this dataset and noticed that vehicular honks are well recognized with 96.85% accuracy. The accuracy falls slightly because few noise samples like drilling are detected as a honk. This result shows that our proposed model provides satisfactory accuracy even in different circumstances.

6.4. Impact of different smartphone models

In our next experiment, the performance of smartphones is simulated by measuring the number of honks. We have used three smartphones (Samsung Galaxy M30s, Motorola G40 Fusion, and RealMe C15) to estimate the accuracy of the sound sensors in honks identification. These phones vary in terms of cost too. To estimate the accuracy, we captured a sample vehicular honk data in road traffic simultaneously by starting the recording of all the smartphones at the same time and same location for the duration of 6 min. Data were tested with ResNet152 model and Fig. 14 shows that the smartphones with their accuracy where the accuracy level among them is varied less than 5%. The result suggests that different smartphones might be considered in measuring honks in our work.

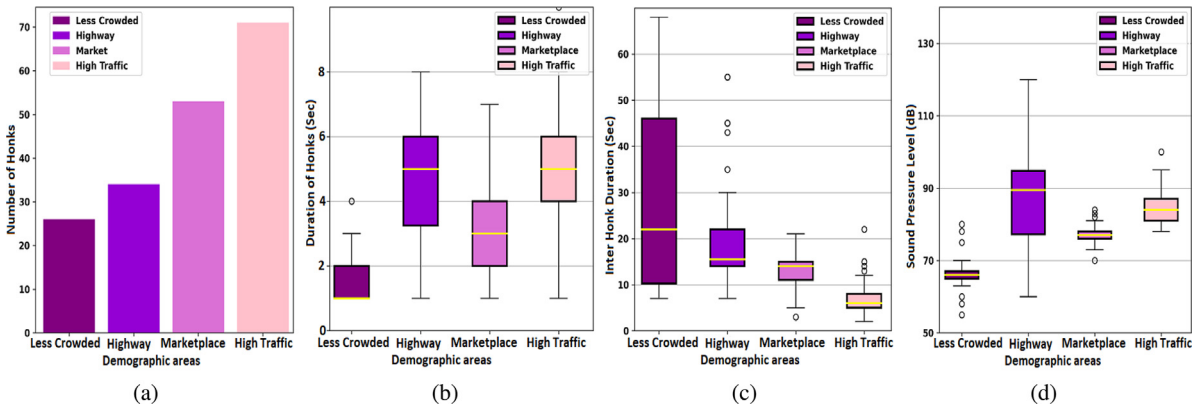


Fig. 15. Distribution of (a) Number of honks (b) Duration of honks (c) Inter honk duration, and (d) Sound pressure level of our dataset.

6.5. Honk characteristics and analysis

Once the honk identification is determined using the ResNet152 model, we can retrieve different honk features as stated earlier for each audio sample. However, in Section 4.1, we have reviewed the honk features where the samples were manually tagged. In this subsection, we analyze how our proposed system generates the honk features of the audio file to identify the possible context of a location. Here, four different locations (less crowded area, highway, marketplace, high traffic area) are considered, and verified the distribution of each feature. Fig. 15(a) demonstrates the number of honks, where we can see that the high traffic area contains a maximum number of honks. The box plot of the distribution of each honk duration is shown in Fig. 15(b). It is noticed that this distribution is almost the same in highway and high traffic areas, while it is comparatively low in less crowded places. However, an opposite tendency has been observed in Fig. 15(c). The figure shows the box plot of inter honk duration between two successive honks, where low traffic area has the maximum inter honk duration and high traffic area possesses the minimum inter honk duration. Fig. 15(d) represents the variation of sound pressure level. The nature of duration of honks and SPL is almost same, but the distribution is different for both the cases. From Fig. 15, it is found that a single feature is not enough for context identification. Every feature is correlated with each other, and when we combine all three features, a unique identity can be established, and that definitely qualifies to predict the context of a location.

Context Interpretation: In this study, we have highlighted the demographic characteristics based on honking features by moving through four different areas in a single trace to make an intuitive basis for the perceived result. Fig. 16 delineates the honking features of a residential area, normal traffic area, marketplace, and high traffic area. From the observations as shown in Fig. 15, it is almost verified with the scrutiny of the physical location. Here, we have considered the morning slot for ground truth verification and seen that the number of honks, duration of honks, the gap between two successive honks, and SPL clearly justify the context of a location.

6.6. Discussion & remarks

- **Significance of dataset**: Results show that the performance of best-performing baseline model (CNN [16]) increases by 18.29% accuracy when trained using our (noisy) dataset in contrast to the existing (noise-free) UrbanSound8K dataset. The proposed ResNet152 model also shows similar trends with 11% enhanced accuracy when trained using our dataset compared to that of UrbanSound8k. This observation reveals the importance of a good training dataset on model performance irrespective of the choice of models.
- **Importance of our proposed model**: Our study shows that the best transfer learning based model outperforms all existing variants of traditional CNN models. The main reason is that the transfer learning models are already trained with sufficiently large volumes of image repository datasets. Such pre-training can be properly exploited to achieve significant performance gains when audio samples are modeled as a spectrogram image. The proposed model is found to successfully filter out all non-honk audio samples with very high accuracy. Hence it can be used effectively to detect honk from audio streams.
- **Sensing honk in IoT edge device**: Performance accuracy is obviously an important metric in choosing models. However, there may be scenarios where the detection needs to be in real time, and the models also need to be executed in some resource constrained devices like smartphones or IoT edge devices. In such case, we recommend the use of the MobileNet model, which is relatively lightweight and without significant degradation of performance approx. 6% as observed in our experiment.



Fig. 16. Ground truth verification of honk features with the physical location.

- **Context-based inference**: Precise detection of honk features enable the detection of outdoor context. It has got a significant impact in exploring passive sensing techniques to identify traffic congestion, traffic density estimation, pollution exposures etc.
  In the next section we discuss about a case study of route recommendation micro-service.

7. Micro-service: Route recommendation

The study discussed in Section 6.5 helps to identify the Fh_L values, which further facilitates us to design a honk-aware route recommendation system. To do that, we collected data from a sub-urban town in West Bengal, India. We chose three different routes for the same source & destination in that town. Continuous 20 days of data were collected for those three routes in three different time slots. The reason behind choosing a different town for route recommendation is to check how well our proposed framework works in a different place. In the following subsections, we describe the steps to find a honk-aware route recommendation system.

7.1. Cluster formation

K-means algorithm [47] is used to form the clusters based on the number of honks, duration of honks, inter gap between consecutive honks, and SPL level. To determine the value of k to perform the K-means clustering algorithm, the elbow method is used, which provides the value of $k = 3$ in our case as depicted in Fig. 17. Since we have used four different features as an input of the K-means algorithm, to show the output in a 2-dimension plot, we have further applied nonlinear dimensionality reduction technique (tSNE) to reduce our four dimensions data into two dimensions. Fig. 17 represents three different clusters based on our dataset.

7.2. Model learning

A multidimensional feature space is generated after summing up all the cluster values. Therefore, a machine learning-based approach is mandated to train the system when the feature value is unknown to us. In this research, we have

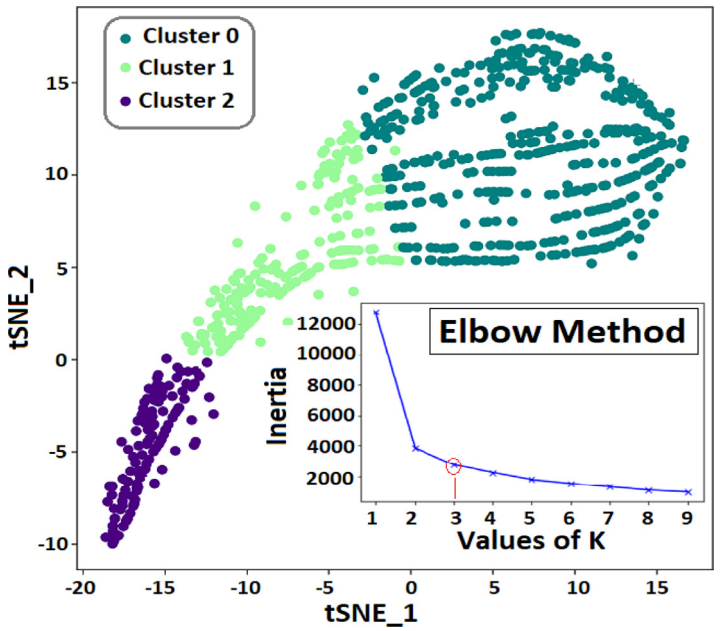


Fig. 17. Illustration of elbow method and K-means algorithm using tSNE for feature reduction.

Table 11
Evaluation result of different machine learning models.

	LR	RF	XGB	SVM
Accuracy	98.52%	99.26%	98.52%	99.26%
Precision	0.96	0.98	0.96	0.99
Recall	0.97	0.99	0.99	0.98
F1-Score	0.96	0.99	0.97	0.99
MCC Score	0.97	0.98	0.97	0.99
Kappa statistics	0.97	0.98	0.97	0.99

considered Logistic Regression (LR) [48], Random Forest Classifier (RF) [48], XGB classifier (XGB) [49], and Support Vector Machine (SVM) [48] models to determine the appropriate clusters for our dataset. A total of 1352 samples is considered by collecting three different routes of the same source and destination. We have collected twenty days of continuous data for three different time slots (i.e., morning, afternoon, and evening). Here, 1081 samples are used for training, and 271 samples are used for testing. The performance of each model is shown in Table 11. We can see that both RF and SVM models give an accuracy of 99.26% which is better than the accuracy obtained by LR and XGB models. With respect to other metrics such as Precision, MCC Score, and Kappa statistics, the SVM model performs slightly better than the RF model. Hence, we have considered the SVM model in our work for further analysis.

7.3. Route recommendation

Among the three different clusters, we have noticed that Cluster 0 is highly polluted cluster, whereas Cluster 2 is less polluted cluster. Hence, Cluster 0 is most harmful and Cluster 2 is less harmful as compared to other clusters. In our work, each cluster is assigned some weight values based on the severity of human health. We assign weight values 3 to Cluster 0, 2 to Cluster 1, and 1 to Cluster 0. At first, we calculate the total number of each cluster for each route (see Eq. (1)) at a given time. After that, the proportion of each cluster is estimated using Eq. (2), which provides some idea about the engagement of each cluster in a particular route at a specific time.

$$R_i(C_j) = f(C_j) \tag{1}$$

where C_j represents each cluster, $f(c_j)$ = Frequency of occurrence of cluster C_j in ith route, and $R_i(C_j)$ = number of each cluster in ith route.

$$P(R_i(C_j)) = \frac{R_i(C_j)}{TC(R_i)} \tag{2}$$

where $TC(R_i)$ = Total number of clusters in ith route, $P(R_i(C_j))$ = Proportion of each cluster in ith route.

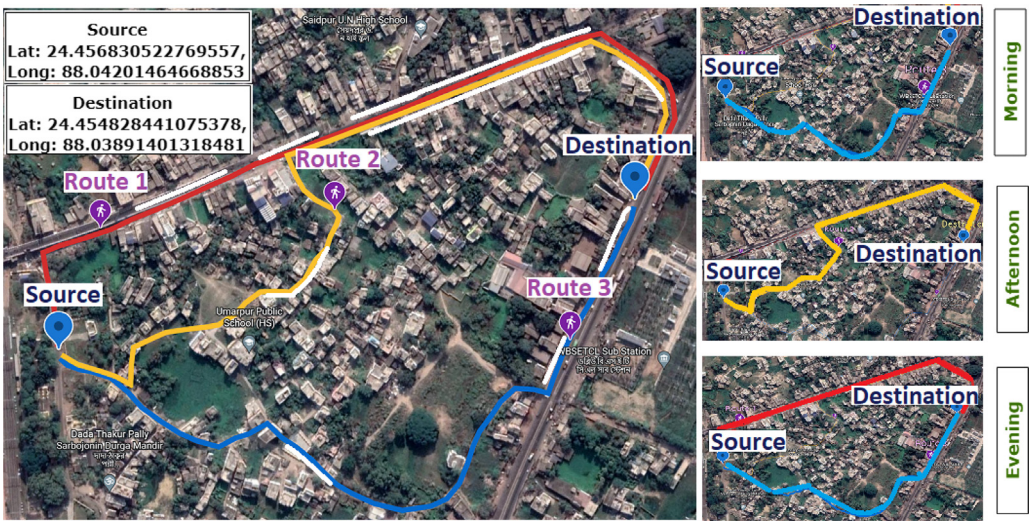


Fig. 18. Actual route along with our recommended route for three different time.

Table 12
Exploration of honk-aware Route.

	Morning			Afternoon			Evening		
	Route 1	Route 2	Route 3	Route 1	Route 2	Route 3	Route 1	Route 2	Route 3
Cluster 0/Proportion (Highly polluted)	23/0.53	31/0.72	20/0.42	21/0.58	32/0.65	31/0.57	20/0.47	30/0.71	29/0.59
Cluster 1/Proportion (Moderately polluted)	19/0.44	6/0.14	21/0.44	16/0.43	9/0.18	21/0.39	23/0.53	3/0.07	14/0.29
Cluster 2/Proportion (Less polluted)	1/0.02	6/0.14	7/0.15	0/0	8/0.16	2/0.04	0/0	9/0.21	6/0.12
Pollution level	2.49	2.58	**2.29**	2.6	**2.47**	2.53	**2.47**	2.48	**2.47**
Healthier	**Route 3**			**Route 2**			**Route 1/Route 3**		

Next, we multiply the weight of each cluster by the proportion of the respective cluster of a route and add them in order to measure the pollution level of that route as shown in Eq. (3). Finally, the minimum polluted route is selected as the best travel route for a source–destination pair (see Eq. (4)).

$$PL(R_i) = \sum_{k=0}^{n} P(R_i(C_{(j+k)\%n})) * w_k \tag{3}$$

where $w_0 > w_1 > w_2 > \cdots > w_n$, $PL(R_i) =$ *Pollution level of each route*, $w_i =$ *weight of* C_j.

$$Route = minimum(PL(R_i)) \tag{4}$$

In this paper, we present an example to illustrate our proposed honk-aware route recommendation system to recognize the healthier route by avoiding noise pollution. As per the user experience, three different routes for the same source and destination are adopted, which is shown in Fig. 18. The Red line denotes $Route_1$, the yellow line shows $Route_2$, the blue line for $Route_3$, and the white line signifies the noise polluted zone of these routes. As per Eqs. (1) to (4), Table 12 is tabulated. As shown in the table, in the morning slot, $Route_3$ contains the lowest pollution level (2.29 units) compared to others, which is 8% and 11.24% less polluted than $Route_1$, and $Route_2$. In case of the afternoon slot, $Route_2$ displays the minimum pollution exposure i.e., 2.47 units, and it is 5%, 2.28% less polluted than $Route_1$, $Route_3$. In the evening slot, $Route_1$, and $Route_3$ are selected because the pollution level is slightly lower than $Route_2$.

With this example, we can conclude that only the vehicular honk and its features are sufficient enough to identify the honk-aware route by avoiding environmental noise pollution.

8. Conclusion & future scope

The repellent features of honking in road traffic create several repercussions on daily human life. Hence, it is important to identify honks from traffic noise. In this paper, we have presented a novel framework De$\mathcal{H}$onk that estimates the number of honks in road traffic in the presence of ambient noise and assesses the different features of honks like duration of honks and inter honk duration between two successive honks, which can further be used to predict the nature of a location and to design several microservices. To measure the number of honks, the collected raw samples are

converted into spectrograms and fed them different deep learning models. Extensive experiments have been performed with respect to various performance indicators, and the results show that the ResNet152 provides the best accuracy in honk identification. Furthermore, we studied various intuitive features of honks to find the spatio-temporal characteristics between them. We have also done the ground truth verification with a physical location based on the feature values.

In the future, we will try to address the following: (1) To improve the accuracy, we can also incorporate the geographic characteristics and our honk data to identify the context of a location and to design some micro-services such as driver rating, smart city planning, traffic management, etc., and (2) Use of the honk profile to estimate the traffic air pollution can be other interesting aspects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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